# Linear Algebra — Model Test 2: Answers and Methods

Covers Sections 1.1–1.3 and 2.1–2.6 · Level: moderate

## Key methods used in this test

- **Linear combinations (1.1):** $cv + dw = b$ is a system of equations in $c, d$.
- **Inequalities (1.2):** triangle inequality $\|x + y\| \leq \|x\| + \|y\|$; $\|x - y\|^2 = \|x\|^2 - 2x \cdot y + \|y\|^2$.
- **Powers of a matrix (2.4):** write $A = I + N$ with $N^3 = 0$; since $IN = NI$ the binomial theorem applies.
- **Elimination matrices (2.3):** $E_{ij}$ is $I$ with $-\ell_{ij}$ in position $(i, j)$; it subtracts $\ell_{ij}\times$ row $j$ from row $i$.
- **Inverse (2.5):** $(Q \,|\, I) \to (I \,|\, Q^{-1})$; then $x = Q^{-1}b$.
- $A = LU$ **(2.6):** $L$ stores the multipliers. To solve $Ax = b$: forward-substitute $Lc = b$, then back-substitute $Ux = c$.

## 1 Vectors

**(a)**

**Method.** Write $cv + dw = b$ component by component and solve.

$c+2d = 4,\ 2c+d = 5,\ c-d = 1$. From the third, $c = 1+d$; substituting in the first, $1+3d = 4 \Rightarrow d = 1$, $c = 2$. Check the second equation: $2 \cdot 2 + 1 = 5$ ✓

| **Answer.** Yes: $b = 2v + w$. |
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**(b)** $v \cdot w = 2 + 2 - 1 = 3$, $\|v\| = \|w\| = \sqrt{6}$, so $\cos\theta = \frac{3}{6} = \frac{1}{2}$.

| **Answer.** $\theta = 60^\circ = \pi/3$. |
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**(c)**

**Method.** Use $\|x - y\|^2 = \|x\|^2 + \|y\|^2 - 2x \cdot y = 25 - 2x \cdot y$ and $-\|x\|\|y\| \leq x \cdot y \leq \|x\|\|y\|$ (Schwarz inequality).

Since $-12 \leq x \cdot y \leq 12$: $1 \leq \|x - y\|^2 \leq 49$. The largest value 7 occurs when $y$ points opposite to $x$; the smallest value 1 when $y$ points the same way as $x$. If $x \perp y$ then $x \cdot y = 0$ and $\|x - y\| = \sqrt{25} = 5$.

| **Answer.** Largest 7, smallest 1; if $x \perp y$ then $\|x - y\| = 5$. |
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## 2 Matrix operations

**(1)**

**Method.** Compute $AB$ $(2 \times 3)$ first, then multiply by $C$.

$$AB = \begin{pmatrix} 6 & 5 & -2 \\ -1 & 9 & 11 \end{pmatrix}, \qquad ABC = \begin{pmatrix} 6 & 5 & -2 \\ -1 & 9 & 11 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ 0 & 1 \\ 3 & 2 \end{pmatrix} = \begin{pmatrix} 0 & -11 \\ 32 & 33 \end{pmatrix}.$$

**Answer.** $ABC = \begin{pmatrix} 0 & -11 \\ 32 & 33 \end{pmatrix}$

**(2)(a)**

**Method.** Split $A = I + N$ with $N$ strictly upper triangular.

$$N = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad N^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad N^3 = 0.$$

Since $IN = NI$: $A^n = (I+N)^n = I + nN + \binom{n}{2}N^2 = \begin{pmatrix} 1 & n & \frac{n(n-1)}{2} \\ 0 & 1 & n \\ 0 & 0 & 1 \end{pmatrix}$. For $n = 10$, $\binom{10}{2} = 45$.

**Answer.** $A^{10} = \begin{pmatrix} 1 & 10 & 45 \\ 0 & 1 & 10 \\ 0 & 0 & 1 \end{pmatrix}$

**(b)**

**Method.** Factor $f(x) = x(x-1)^2 + 3$ to use $N^3 = 0$.

$f(A) = A(A-I)^2 + 3I = (I+N)N^2 + 3I = N^2 + 3I$.

**Answer.** $f(A) = \begin{pmatrix} 3 & 0 & 1 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}$

## 3 Elimination matrices

**(a)**

**Method.** $\ell_{21} = 3/1 = 3$, $\ell_{31} = 0$. After the first step, $\ell_{32} = 4/2 = 2$.

$$E_{21} = \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \; E_{21}A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{pmatrix}, \; E_{32} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{pmatrix}, \; U = E_{32}E_{21}A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{pmatrix}.$$

**(b)** $E = E_{32}E_{21} = \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 6 & -2 & 1 \end{pmatrix}$. (Note the entry $6 = (-2)(-3)$: $E$ is not simply the multipliers with signs changed. That is why $L = E^{-1}$ is easier to work with.)

**(c)** Apply the same steps to $b$: $c = Eb = \begin{pmatrix} 2 \\ 6 \\ -10 \end{pmatrix}$. Back-substitute in $Ux = c$: $5x_3 = -10 \Rightarrow x_3 = -2$; $2x_2 - 2x_3 = 6 \Rightarrow x_2 = 1$; $x_1 + 2x_2 + x_3 = 2 \Rightarrow x_1 = 2$.

**Answer.** $U = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{pmatrix}$, $E = \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 6 & -2 & 1 \end{pmatrix}$, $x = (2, 1, -2)^T$.

## 4 Inverse matrix

**(1)**

**Method.** Row-reduce $(Q \,|\, I)$.

$$\left(\begin{array}{ccc|ccc} 1 & 3 & -2 & 1 & 0 & 0 \\ 1 & 1 & -1 & 0 & 1 & 0 \\ -2 & -1 & 2 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_2 - R_1,\ R_3 + 2R_1} \left(\begin{array}{ccc|ccc} 1 & 3 & -2 & 1 & 0 & 0 \\ 0 & -2 & 1 & -1 & 1 & 0 \\ 0 & 5 & -2 & 2 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_3 + 2R_2,\ R_2 \leftrightarrow R_3,\ R_1 - 3R_2,\ R_3 + 2R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & -6 & -3 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 0 & 0 & 1 & -1 & 5 & 2 \end{array}\right)$$

$$\xrightarrow{R_1 + 2R_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 4 & 1 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 0 & 0 & 1 & -1 & 5 & 2 \end{array}\right)$$

**Answer.** $Q^{-1} = \begin{pmatrix} -1 & 4 & 1 \\ 0 & 2 & 1 \\ -1 & 5 & 2 \end{pmatrix}$

**(2)** $x = Q^{-1}b = \begin{pmatrix} -1 & 4 & 1 \\ 0 & 2 & 1 \\ -1 & 5 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 10 \\ 7 \\ 15 \end{pmatrix}$. Check: $Qx = (10+21-30,\ 10+7-15,\ -20-7+30)^T = (1, 2, 3)^T$ ✓

**Answer.** $x = (10, 7, 15)^T$

## 5 $A = LU$

**(a)**

**Method.** Eliminate without row exchanges, recording multipliers.

$$\begin{pmatrix} 2 & 4 & -2 \\ 4 & 9 & -3 \\ -2 & -3 & 7 \end{pmatrix}$$

$$\xrightarrow{R_2-2R_1,\ R_3+R_1} \begin{pmatrix} 2 & 4 & -2 \\ 0 & 1 & 1 \\ 0 & 1 & 5 \end{pmatrix}$$

$$\xrightarrow{R_3-R_2} \begin{pmatrix} 2 & 4 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 4 \end{pmatrix}$$

$\ell_{21} = 2,\ \ell_{31} = -1,\ \ell_{32} = 1$: $L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{pmatrix},\ U = \begin{pmatrix} 2 & 4 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 4 \end{pmatrix}$.

**(b)**

**Method.** Forward substitution $Lc = b$, then back substitution $Ux = c$.

$Lc = b$: $c_1 = 2$, $c_2 = 8 - 2c_1 = 4$, $c_3 = 10 + c_1 - c_2 = 8$. $Ux = c$: $4x_3 = 8 \Rightarrow x_3 = 2$; $x_2 + x_3 = 4 \Rightarrow x_2 = 2$; $2x_1 + 4x_2 - 2x_3 = 2 \Rightarrow x_1 = -1$.
Check: $Ax = (-2 + 8 - 4,\ -4 + 18 - 6,\ 2 - 6 + 14) = (2, 8, 10)$ ✓

**Answer.** $L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{pmatrix}$, $U = \begin{pmatrix} 2 & 4 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 4 \end{pmatrix}$; $x = (-1, 2, 2)^T$.